山东大学 计算机科学与技术 学院

自然语言处理 课程实验报告

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实验题目： 实验三-词性标注

实验内容：

1.利用 Chinese.txt 和 English.txt 的中英文句子，在实验二的基础上，继续利用以下给定的中英文工具进行词性标注。并对不同工具产生的结果进行简要对比分析，将实验过程与结果写成实验报告，实验课结束后提交。

2.使用 python 编程实践 CRF，进行词性标注。该实验基于 python3.6 以及 keras 训练 bi-lstm,结合 CRF 来实现词性标注，本实验可以在 ModelArts(华为云)平台进行，也可以在在本机进行。

实验过程中遇到和解决的问题：

（记录实验过程中遇到的问题，以及解决过程和实验结果。可以适当配以关键代码辅助说明，但不要大段贴代码。）

实验遇到的问题：

实验材料没有提供 CRF 训练需要的训练集

解决措施：网络搜索，使用人民日报数据集当作中文训练的训练集，使用 https://github.com/UniversalDependencies/UD_English-EWT/blob/master/not-to-release/sources/email/enronsent00_01.xml.conllu 下的 universaldependencies 中的部分数据作为训练集。

实验步骤：

实验中 StanfordCoreNLP 更换使用为升级版的 Stanza

一、在实验二基础上进行词性标注

(1) English.txt 词性标注

调用 NLTK，Spacy，Stanza 库进行词性标注

(2) Chinese.txt

调用 Jieba，Stanza，SnowNLP，THULAC，NLPIR 库进行词性标注

二、编程实践 CRF 进行词性标注

（1）将训练集读取并预处理为 sentences 列表和对应 tag 列表，如下格式

```
sentences = [
    ["我", "喜欢", "吃", "苹果"],
    ["他", "正在", "学习", "中文"]
    ]
tags = [
    ["r", "v", "v", "n"],
    ["r", "d", "v", "n"]
]
```

（2）采用 BiLSTM-CRF 构架，包含输入层接受固定长度的序列输入；嵌入层将词语索引映射为密集向量，双向 LSTM 层用以捕捉上下文信息，全连接层输出每个位置的标签得

分，CRF 层：建模标签间的转移关系。

实验结果：

一、在实验二的基础上进行词性标注

(1) English.txt 词性标注

使用 NLTK 进行词性标注的结果：

```
(nlp) D:\大学\大三下\自然语言处理\实验\exp3>python en.py
NLTK标注结果: [('Xi', 'NN'), ('Jinping', 'NNP'), (',', ','), ('male', 'NN'), (',', ','), ('Han', 'NNP'), ('ethnicity', 'NN'), (',', ','), ('was', 'VBD'), ('born', 'VBN'), ('in', 'IN'), ('June', 'NNP'), ('1953', 'CD'), ('and', 'CC'), ('is', 'VBZ'), ('from', 'IN'), ('Fuping', 'VBG'), (',', ','), ('Shaanxi', 'NNP'), ('Province', 'NNP'), ('.', '.'), ('He', 'PRP'), ('began', 'VBD'), ('his', 'PRP$'), ('first', 'JJ'), ('job', 'NN'), ('in', 'IN'), ('January', 'NNP'), ('1969', 'CD'), ('and', 'CC'), ('joined', 'VBD'), ('the', 'DT'), ('Communist', 'NNP'), ('Party', 'NNP'), ('of', 'IN'), ('China', 'NNP'), ('(', '('), ('CPC', 'NNP'), (')', ')'), ('in', 'IN'), ('January', 'NNP'), ('1974', 'CD'), ('.', '.'), ('Xi', 'VB'), ('graduated', 'VBN'), ('from', 'IN'), ('School', 'NNP'), ('of', 'IN'), ('Humanities', 'NNP'), ('and', 'CC'), ('Social', 'NNP'), ('Sciences', 'NNPS'), (',', ','), ('Tsinghua', 'NNP'), ('University', 'NNP'), ('where', 'WRB'), ('he', 'PRP'), ('completed', 'VBD'), ('an', 'DT'), ('in-service', 'JJ'), ('graduate', 'NN'), ('program', 'NN'), ('in', 'IN'), ('Marxist', 'NNP'), ('theory', 'NN'), ('and', 'CC'), ('ideological', 'JJ'), ('and', 'CC'), ('political', 'JJ'), ('education', 'NN'), ('.', '.'), ('He', 'PRP'), ('holds', 'VBZ'), ('a', 'DT'), ('Doctor', 'NNP'), ('of', 'IN'), ('Law', 'NNP'), ('degree', 'NN'), ('.', '.'), ('Xi', 'NN'), ('is', 'VBZ'), ('currently', 'RB'), ('General', 'NNP'), ('Secretary', 'NNP'), ('of', 'IN'), ('the', 'DT'), ('CPC', 'NNP'), ('Central', 'NNP'), ('Committee', 'NNP'), (',', ','), ('Chairman', 'NNP'), ('of', 'IN'), ('the', 'DT'), ('CPC', 'NNP'), ('Central', 'NNP'), ('Military', 'NNP'), ('Commission', 'NNP'), (',', ','), ('President', 'NNP'), ('of', 'IN'), ('the', 'DT'), ('People', 'NNP'), ("'s", 'POS'), ('Republic', 'NNP'), ('of', 'IN'), ('China', 'NNP'), ('(', '('), ('PRC', 'NNP'), (')', ')'), (',', ','), ('and', 'CC'), ('Chairman', 'NNP'), ('of', 'IN'), ('the', 'DT'), ('PRC', 'NNP'), ('Central', 'NNP'), ('Military', 'NNP'), ('Commission', 'NNP'), ('.', '.')]
```

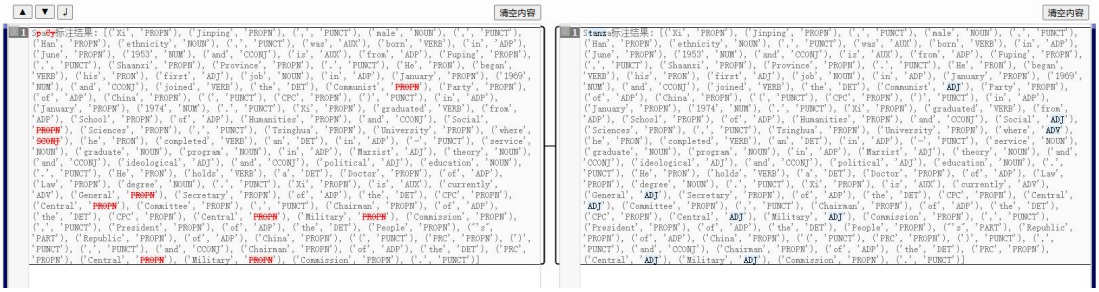
使用 SpaCy 进行词性标注的结果：

```
SpaCy标注结果: [('Xi', 'PROPN'), ('Jinping', 'PROPN'), (',', 'PUNCT'), ('male', 'NOUN'), (',', 'PUNCT'), ('Han', 'PROPN'), ('ethnicity', 'NOUN'), (',', 'PUNCT'), ('was', 'AUX'), ('born', 'VERB'), ('in', 'ADP'), ('June', 'PROPN'), ('1953', 'NUM'), ('and', 'CCONJ'), ('is', 'AUX'), ('from', 'ADP'), ('Fuping', 'PROPN'), (',', 'PUNCT'), ('Shaanxi', 'PROPN'), ('Province', 'PROPN'), ('.', 'PUNCT'), ('He', 'PRON'), ('began', 'VERB'), ('his', 'PRON'), ('first', 'ADJ'), ('job', 'NOUN'), ('in', 'ADP'), ('January', 'PROPN'), ('1969', 'NUM'), ('and', 'CCONJ'), ('joined', 'VERB'), ('the', 'DET'), ('Communist', 'PROPN'), ('Party', 'PROPN'), ('of', 'ADP'), ('China', 'PROPN'), ('(', 'PUNCT'), ('CPC', 'PROPN'), (')', 'PUNCT'), ('in', 'ADP'), ('January', 'PROPN'), ('1974', 'NUM'), ('.', 'PUNCT'), ('Xi', 'PROPN'), ('graduated', 'VERB'), ('from', 'ADP'), ('School', 'PROPN'), ('of', 'ADP'), ('Humanities', 'PROPN'), ('and', 'CCONJ'), ('Social', 'PROPN'), ('Sciences', 'PROPN'), (',', 'PUNCT'), ('Tsinghua', 'PROPN'), ('University', 'PROPN'), ('where', 'SCONJ'), ('he', 'PRON'), ('completed', 'VERB'), ('an', 'DET'), ('in', 'ADP'), ('-', 'PUNCT'), ('service', 'NOUN'), ('graduate', 'NOUN'), ('program', 'NOUN'), ('in', 'ADP'), ('Marxist', 'ADJ'), ('theory', 'NOUN'), ('and', 'CCONJ'), ('ideological', 'ADJ'), ('and', 'CCONJ'), ('political', 'ADJ'), ('education', 'NOUN'), ('.', 'PUNCT'), ('He', 'PRON'), ('holds', 'VERB'), ('a', 'DET'), ('Doctor', 'PROPN'), ('of', 'ADP'), ('Law', 'PROPN'), ('degree', 'NOUN'), ('.', 'PUNCT'), ('Xi', 'PROPN'), ('is', 'AUX'), ('currently', 'ADV'), ('General', 'PROPN'), ('Secretary', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('CPC', 'PROPN'), ('Central', 'PROPN'), ('Committee', 'PROPN'), (',', 'PUNCT'), ('Chairman', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('CPC', 'PROPN'), ('Central', 'PROPN'), ('Military', 'PROPN'), ('Commission', 'PROPN'), (',', 'PUNCT'), ('President', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('People', 'PROPN'), ("'s", 'PART'), ('Republic', 'PROPN'), ('of', 'ADP'), ('China', 'PROPN'), ('(', 'PUNCT'), ('PRC', 'PROPN'), (')', 'PUNCT'), (',', 'PUNCT'), ('and', 'CCONJ'), ('Chairman', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('PRC', 'PROPN'), ('Central', 'PROPN'), ('Military', 'PROPN'), ('Commission', 'PROPN'), ('.', 'PUNCT')]
2025-04-19 21:10:00 WARNING: Language en package default expects mwt, which has been added
```

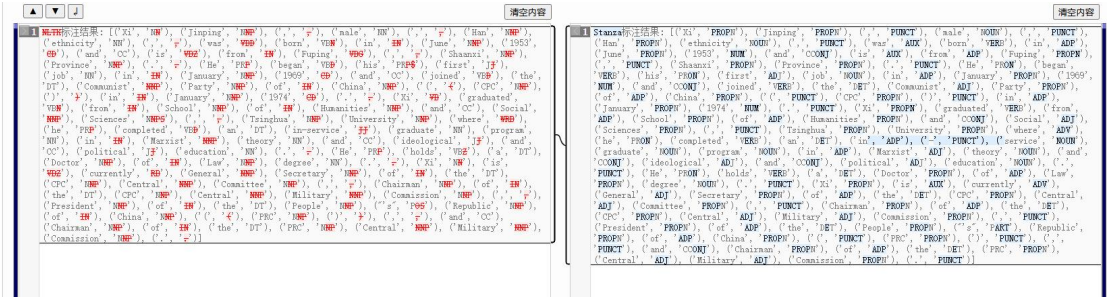
使用 Stanza 词性标注的结果：

```
Stanza标注结果: [('Xi', 'PROPN'), ('Jinping', 'PROPN'), (',', 'PUNCT'), ('male', 'NOUN'), (',', 'PUNCT'), ('Han', 'PROPN'), ('ethnicity', 'NOUN'), (',', 'PUNCT'), ('was', 'AUX'), ('born', 'VERB'), ('in', 'ADP'), ('June', 'PROPN'), ('1953', 'NUM'), ('and', 'CCONJ'), ('is', 'AUX'), ('from', 'ADP'), ('Fuping', 'PROPN'), (',', 'PUNCT'), ('Shaanxi', 'PROPN'), ('Province', 'PROPN'), ('.', 'PUNCT'), ('He', 'PRON'), ('began', 'VERB'), ('his', 'PRON'), ('first', 'ADJ'), ('job', 'NOUN'), ('in', 'ADP'), ('January', 'PROPN'), ('1969', 'NUM'), ('and', 'CCONJ'), ('joined', 'VERB'), ('the', 'DET'), ('Communist', 'ADJ'), ('Party', 'PROPN'), ('of', 'ADP'), ('China', 'PROPN'), ('(', 'PUNCT'), ('CPC', 'PROPN'), (')', 'PUNCT'), ('in', 'ADP'), ('January', 'PROPN'), ('1974', 'NUM'), ('.', 'PUNCT'), ('Xi', 'PROPN'), ('graduated', 'VERB'), ('from', 'ADP'), ('School', 'PROPN'), ('of', 'ADP'), ('Humanities', 'PROPN'), ('and', 'CCONJ'), ('Social', 'ADJ'), ('Sciences', 'PROPN'), (',', 'PUNCT'), ('Tsinghua', 'PROPN'), ('University', 'PROPN'), ('where', 'ADV'), ('he', 'PRON'), ('completed', 'VERB'), ('an', 'DET'), ('in', 'ADP'), ('-', 'PUNCT'), ('service', 'NOUN'), ('graduate', 'NOUN'), ('program', 'NOUN'), ('in', 'ADP'), ('Marxist', 'ADJ'), ('theory', 'NOUN'), ('and', 'CCONJ'), ('ideological', 'ADJ'), ('and', 'CCONJ'), ('political', 'ADJ'), ('education', 'NOUN'), ('.', 'PUNCT'), ('He', 'PRON'), ('holds', 'VERB'), ('a', 'DET'), ('Doctor', 'PROPN'), ('of', 'ADP'), ('Law', 'PROPN'), ('degree', 'NOUN'), ('.', 'PUNCT'), ('Xi', 'PROPN'), ('is', 'AUX'), ('currently', 'ADV'), ('General', 'ADJ'), ('Secretary', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('CPC', 'PROPN'), ('Central', 'ADJ'), ('Committee', 'PROPN'), (',', 'PUNCT'), ('Chairman', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('CPC', 'PROPN'), ('Central', 'ADJ'), ('Military', 'ADJ'), ('Commission', 'PROPN'), (',', 'PUNCT'), ('President', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('People', 'PROPN'), ("'s", 'PART'), ('Republic', 'PROPN'), ('of', 'ADP'), ('China', 'PROPN'), ('(', 'PUNCT'), ('PRC', 'PROPN'), (')', 'PUNCT'), (',', 'PUNCT'), ('and', 'CCONJ'), ('Chairman', 'PROPN'), ('of', 'ADP'), ('the', 'DET'), ('PRC', 'PROPN'), ('Central', 'ADJ'), ('Military', 'ADJ'), ('Commission', 'PROPN'), ('.', 'PUNCT')]
```

简要分析：



使用文本对比工具进行对比，SpaCy 词性标注与 Stanza 词性标注大部分相同，只有少部分如”Communist”, “Social”,”where”, ”General”, “Central”等词语标注不同



对比 NLTK 与 Stanza，两者词性标注不同较多，其中一部分原因是两者形容同一个词性的词语不同，另一个原因是有些词语的确被不同标注。

(2) Chinese.txt

使用 jieba 进行词性标注结果：

```
=== Jieba 标注结果 ===
Building prefix dict from the default dictionary ...
Loading model from cache C:\Users\薛润泽\AppData\Local\Temp\jieba.cache
Loading model cost 0.558 seconds.
Prefix dict has been built successfully.
2025-04-19 20:47:04 INFO: "zh" is an alias for "zh-hans"
2025-04-19 20:47:04 INFO: Loading these models for language: zh-hans (Simplified_Chinese):
=============================
| Processor | Package       |
-----------------------------
| tokenize  | gsdsimp       |
| pos       | gsdsimp_charlm |
=============================

2025-04-19 20:47:04 INFO: Using device: cpu
2025-04-19 20:47:04 INFO: Loading: tokenize
2025-04-19 20:47:04 INFO: Loading: pos
2025-04-19 20:47:04 INFO: Done loading processors!
3/m 月/m 23/m 日/m 下午/t ，/x "/x 青年报/n 杯赛/n "/x U19/eng 邀请赛/n 在/p 越南/ns 芽庄/n 进行/v ，/x 前/f 国脚/n 曲波/nr 挂帅/n 的/uj 中国/ns U19B/eng 队/n 迎战/v 泰国/ns U19/eng ，/x 上半场/ns 国青队/nt 的/uj 门户/n 大/a 开/v ，/x 泰国/ns 在/p 第/m 11/m 分钟/q 和/c 第/m 17/m 分钟/q 连进/a 2/m 球/n ，/x 半场/n 国青/nt 0/m 射门/n 0/x 角球/n ，/x 几乎/d 被/p 完全/ad 压制/n 。/x 下半场/n ，/x 国青/nr 的/uj 进攻/v 一度/mq 有所/n 起色/n ，/x 并/c 由/p 马辅逊/nr 利用/n 远射/v 扳回/v 一/m 球/n ，/x 但/c 最终/d 未能/v 扳平/n 比分/d 。/x 全场/n 比赛/vn 结束/v ，/x 国青/nt 1/m -/x 2/x 输球/n ，/x 继/v 中国/ns 杯国/n 足/a 0/m -/x 1/x 输给/v 泰国/ns 之后/f ，/x 3/m 天内/s 遭遇/n 泰国/ns 足球/n 双/m 杀/v 。/x
```

使用 Stanza 标注结果：

```
=== Stanza 标注结果 ===
3/NUM 月/NOUN 23/NUM 日/NOUN 下午/NOUN ， /PUNCT “/PUNCT 青年/NOUN 报/PART 杯赛/NOUN ”/PUNCT U19/X 邀请/VERB 赛/PART 在/V
ERB 越南/PROPN 芽/PART 庄/PART 进行/VERB ， /PUNCT 前国/NOUN 脚曲/NOUN 波挂/VERB 帅/NOUN 的/PART 中国/PROPN U19B/X 队/PAR
T 迎战/VERB 泰国/PROPN U19/X ， /PUNCT 上/NOUN 半/NUM 场/NOUN 国青/PROPN 队/PART 的/PART 门户/NOUN 大开/VERB ， /PUNCT 泰
国/PROPN 在/ADP 第11/NUM 分钟/NOUN 和/CCONJ 第17/NUM 分钟/NOUN 连进/VERB 2/NUM 球/NOUN ， /PUNCT 半/NUM 场国/NOUN 青0/NUM
 射门/NOUN 0/NUM 角/NOUN 球/PART ， /PUNCT 几乎/ADV 被/AUX 完全/ADV 压制/VERB 。/PUNCT 下半/NUM 场/NOUN ， /PUNCT 国青/PRO
PN 的/PART 进攻/NOUN 一度/ADV 有/VERB 所/ADV 起色/VERB ， /PUNCT 并/ADV 由/VERB 马/PROPN 辅渔/PROPN 利用/VERB 远射/NOUN
扳/VERB 回/VERB 一/NUM 球/NOUN ， /PUNCT 但/ADV 最终/NOUN 未能/AUX 扳平/VERB 比分/NOUN 。/PUNCT 全场/NOUN 比赛/NOUN 结束/
VERB ， /PUNCT 国青/PROPN 1-2/NUM 输球/VERB ， /PUNCT 继/ADP 中国/PROPN 杯/PART 国足/NOUN 0-1/NUM 输/VERB 给/VERB 泰国/PRO
PN 之后/ADP ， /PUNCT 3/NUM 天/NOUN 内/ADP 遭遇/VERB 泰国/PROPN 足球/NOUN 双/NUM 杀/NOUN 。/PUNCT
```

使用 SnowNLP 进行标注的结果：

```
=== SnowNLP 分词结果 ===
[('3', 'ns'), ('月', 'n'), ('23', 'Rg'), ('日', 'Ng'), ('下午', 't'), (', “', 't'), ('青年报', 'n'), ('杯赛', 'Dg'), ('”
U19', 'Ag'), ('邀请赛', 'n'), ('在', 'p'), ('越南', 'ns'), ('芽', 'n'), ('庄', 'nr'), ('进行', 'v'), (', ', 'w'), ('前',
 'f'), ('国', 'n'), ('脚曲', 'nr'), ('波', 'nr'), ('挂帅', 'v'), ('的', 'u'), ('中国', 'ns'), ('U19B', 'nz'), ('队', 'n'
), ('迎战', 'v'), ('泰国', 'ns'), ('U19, ', 'nz'), ('上半场', 'n'), ('国', 'n'), ('青队', 'f'), ('的', 'u'), ('门户', 'n
'), ('大', 'd'), ('开', 'v'), (', ', 'w'), ('泰国', 'ns'), ('在', 'p'), ('第', 'm'), ('11', 'm'), ('分钟', 'q'), ('和',
'c'), ('第', 'm'), ('17', 'm'), ('分钟', 'q'), ('连', 'n'), ('进', 'v'), ('2', 'Bg'), ('球', 'n'), (', ', 'w'), ('半场',
 'nt'), ('国', 'n'), ('青', 'nr'), ('0', 'nr'), ('射门', 'v'), ('0', 'u'), ('角球', 'n'), (', ', 'w'), ('几乎', 'd'), ('
被', 'p'), ('完全', 'ad'), ('压制', 'v'), ('。', 'w'), ('下半场', 'n'), (', ', 'w'), ('国', 'n'), ('青', 'a'), ('的', 'u
'), ('进攻', 'vn'), ('一度', 'd'), ('有所', 'v'), ('起色', 'n'), (', ', 'w'), ('并', 'c'), ('由', 'p'), ('马', 'nr'), ('
辅', 'nr'), ('渔', 'Ng'), ('利用', 'v'), ('远射', 'v'), ('扳回', 'v'), ('一', 'm'), ('球', 'n'), (', ', 'w'), ('但', 'c'
), ('最终', 'd'), ('未', 'd'), ('能', 'v'), ('扳平', 'h'), ('比分', 'n'), ('。', 'w'), ('全场', 'n'), ('比赛', 'vn'), ('
结束', 'v'), (', ', 'w'), ('国', 'n'), ('青', 'nr'), ('1-2', 'nr'), ('输', 'v'), ('球', 'n'), (', ', 'w'), ('继', 'Vg'),
 ('中国', 'ns'), ('杯', 'q'), ('国', 'n'), ('足', 'd'), ('0-1', 'd'), ('输', 'v'), ('给', 'p'), ('泰国', 'ns'), ('之后',
 'f'), (', 3', 'm'), ('天', 'q'), ('内', 'f'), ('遭遇', 'v'), ('泰国', 'ns'), ('足球', 'n'), ('双杀', 'Rg'), ('。', 'w')
]
Model loaded succeed
```

使用 THULAC 进行标注的结果：

```
=== THULAC 标注结果 ===
3月/t 23日/t 下午/t ，/w “/w 青年/n 报杯赛/n ”/w U/g 19/m 邀请赛/n 在/p 越南/ns 芽庄/ns 进行/v ，/w 前/f 国脚/n 曲波/np
挂帅/v 的/u 中国/ns U19B队/n 迎战/v 泰国/ns U19/x ，/w 上半场/n 国青队/ni 的/u 门户/n 大/d 开/v ，/w 泰国/ns 在/p 第11/m
 分钟/q 和/c 第17/m 分钟/q 连/d 进/v 2/m 球/n ，/w 半/m 场/q 国青0/n 射门/v 0/v 角球/n ，/w 几乎/d 被/p 完全/a 压制/v 。
/w 下半场/n ，/w 国青/a 的/u 进攻/v 一度/d 有所/v 起色/n ，/w 并/c 由/p 马辅渔/np 利用/v 远射/v 扳回/v 一/m 球/n ，/w 但
/c 最终/d 未能/v 扳平/v 比分/n 。/w 全场/n 比赛/v 结束/v ，/w 国青/a 1/m -/w 2/m 输/v 球/n ，/w 继/g 中国/ns 杯国/ns 足0
/m -/w 1/m 输/v 给/v 泰国/ns 之后/f ，/w 3/m 天/q 内/f 遭遇/v 泰国/ns 足球/n 双杀/a 。/w
```

使用 NLPIR 进行标注的结果：

```
=== NLPIR 标注结果 ===
3月/time word 23日/time word 下午/time word ， /punctuation mark “/punctuation mark 青年报/multiword expression 杯/noun
赛/verb ”/punctuation mark U19/noun 邀请赛/noun 在/preposition 越南/noun 芽/noun 庄/noun 进行/verb ， /punctuation mark
前/noun of locality 国脚/noun 曲波/noun 挂帅/verb 的/particle 中国/noun U19B/noun 队/noun 迎战/verb 泰国/noun U19/noun
， /punctuation mark 上半场/noun 国青/noun 队/noun 的/particle 门户/noun 大/adverb 开/verb ， /punctuation mark 泰国/noun
在/preposition 第11/numeral 分钟/classifier 和/conjunction 第17/numeral 分钟/classifier 连/verb 进/verb 2/numeral 球/nou
n ， /punctuation mark 半/numeral 场/classifier 国青/noun 0/numeral 射门/verb 0/numeral 角球/noun ， /punctuation mark 几
乎/adverb 被/preposition 完全/adjective 压制/verb 。/punctuation mark 下/verb 半/numeral 场/classifier ， /punctuation ma
rk 国青/noun 的/particle 进攻/verb 一度/adverb 有所/verb 起色/noun ， /punctuation mark 并/conjunction 由/preposition 马/
noun 辅/verb 渔利/verb 用/preposition 远射/verb 扳回/verb 一/numeral 球/noun ， /punctuation mark 但/conjunction 最终/adv
erb 未能/verb 扳平/verb 比分/noun 。/punctuation mark 全场/noun 比赛/verb 结束/verb ， /punctuation mark 国青/noun 1-2/nu
meral 输/verb 球/noun ， /punctuation mark 继/verb 中国/noun 杯/noun 国/noun 足/adjective 0-1/numeral 输/verb 给/preposit
ion 泰国/noun 之后/noun of locality ， /punctuation mark 3/numeral 天/classifier 内/noun of locality 遭遇/verb 泰国/noun
足球/noun 双/numeral 杀/verb 。/punctuation mark
```

简要分析：

Jieba 对常见名词识别比较好，但是对组合词如国青，数字单位结合词如 3 月等词性标注错误，Stanza 标注错误较多，SnowNLP 人名拆分错误较多，THULAC 对专有名词识别好，如 U19B 队，但是对数字和单位分割不自然，NLPIR 分词标注较差。

二、编程实践 CRF 进行词性标注

中文词语词性标注：

（1）

```
(nlp) D:\大学\大三下\自然语言处理\实验\exp3>python test.py
Building prefix dict from the default dictionary ...
Loading model from cache C:\Users\薛润泽\AppData\Local\Temp\jieba.cache
Loading model cost 0.383 seconds.
Prefix dict has been built successfully.
['3', '月', '23', '日', '下午', '青年报', '杯赛', 'U19', '邀请赛', '在', '越南', '芽庄', '进行', '前', '国脚', '曲波', '
挂帅', '的', '中国', 'U19B', '队', '迎战', '泰国', 'U19', '上半场', '国青队', '的', '门户', '大开', '泰国', '在', '第',
11', '分钟', '和', '第', '17', '分钟', '连进', '2', '球', '半场', '国青', '0', '射门', '0', '角球', '几乎', '被', '完全
, '压制', '下半场', '国青', '的', '进攻', '一度', '有所', '起色', '并', '由', '马辅', '渔', '利用', '远射', '扳回', '一
, '球', '但', '最终', '未能', '扳平', '比分', '全场', '比赛', '结束', '国青', '1', '2', '输球', '继', '中国', '杯国足',
'0', '1', '输给', '泰国', '之后', '3', '天内', '遭遇', '泰国', '足球', '双杀']
```

(2)训练 Bi-LSTM 结合 CRF，对分词结果进行词性标注，结果如下：

```
预测结果:
[('3', 'm'), ('月', 'n'), ('23', 'nr'), ('日', 'j'), ('下午', 't'), ('青年报', 'n'), ('杯赛', 'nr'), ('U19', 'ns'), ('邀请
赛', 'n'), ('在', 'p'), ('越南', 'ns'), ('芽庄', 'ns'), ('进行', 'v'), ('前', 'f'), ('国脚', 'ns'), ('曲波', 'n'), ('挂帅',
 'v'), ('的', 'u'), ('中国', 'ns'), ('U19B', 'm'), ('队', 'n'), ('迎战', 'v'), ('泰国', 'ns'), ('U19', 'ns'), ('上半场', 'n
'), ('国青队', 'ns'), ('的', 'u'), ('门户', 'n'), ('大开', 'nr'), ('泰国', 'ns'), ('在', 'p'), ('第', 'ns'), ('11', 'm'), (
'分钟', 'q'), ('和', 'c'), ('第', 'm'), ('17', 'm'), ('分钟', 'q'), ('连进', 'ns'), ('2', 'm'), ('球', 'n'), ('半场', 'n'),
 ('国青', 'm'), ('0', 'n'), ('射门', 'v'), ('0', 'n'), ('角球', 'n'), ('几乎', 'd'), ('被', 'p'), ('完全', 'ad'), ('压制',
'v'), ('下半场', 'n'), ('国青', 'n'), ('的', 'u'), ('进攻', 'vn'), ('一度', 'd'), ('有所', 'v'), ('起色', 'n'), ('并', 'c')
, ('由', 'p'), ('马辅', 'm'), ('渔', 'j'), ('利用', 'v'), ('远射', 'v'), ('扳回', 'v'), ('一', 'm'), ('球', 'n'), ('但', 'c
'), ('最终', 'd'), ('未能', 'v'), ('扳平', 'm'), ('比分', 'n'), ('全场', 'n'), ('比赛', 'vn'), ('结束', 'v'), ('国青', 'm')
, ('1', 'n'), ('2', 'n'), ('输球', 'n'), ('继', 'vg'), ('中国', 'ns'), ('杯国足', 'ns'), ('0', 'ns'), ('1', 'ns'), ('输给',
 'v'), ('泰国', 'ns'), ('之后', 'f'), ('3', 'ns'), ('天内', 'ns'), ('遭遇', 'n'), ('泰国', 'ns'), ('足球', 'n'), ('双杀', '
n')]

(tf) D:\大学\大三下\自然语言处理\实验\exp3>
```

英文词语词性标注：

（1）英文分词结果：

```
D:\大学\大三下\自然语言处理\实验\exp3>conda activate nlp

(nlp) D:\大学\大三下\自然语言处理\实验\exp3>python test.py
[['Xi', 'Jinping', ',', 'male', ',', 'Han', 'ethnicity', ',', 'was', 'born', 'in', 'June', '1953', 'and', 'is', 'from',
'Fuping', ',', 'Shaanxi', 'Province', '.'], ['He', 'began', 'his', 'first', 'job', 'in', 'January', '1969', 'and', 'join
ed', 'the', 'Communist', 'Party', 'of', 'China', '(', 'CPC', ')', 'in', 'January', '1974', '.'], ['Xi', 'graduated', 'fr
om', 'School', 'of', 'Humanities', 'and', 'Social', 'Sciences', ',', 'Tsinghua', 'University', 'where', 'he', 'completed
', 'an', 'in-service', 'graduate', 'program', 'in', 'Marxist', 'theory', 'and', 'ideological', 'and', 'political', 'educ
ation', '.'], ['He', 'holds', 'a', 'Doctor', 'of', 'Law', 'degree', '.'], ['Xi', 'is', 'currently', 'General', 'Secretar
y', 'of', 'the', 'CPC', 'Central', 'Committee', ',', 'Chairman', 'of', 'the', 'CPC', 'Central', 'Military', 'Commission'
, ',', 'President', 'of', 'the', 'People', "'s", 'Republic', 'of', 'China', '(', 'PRC', ')', ',', 'and', 'Chairman', 'of
', 'the', 'PRC', 'Central', 'Military', 'Commission', '.']]
```

（2）训练 Bi-LSTM 结合 CRF，对分词结果进行词性标注，结果如下：

```
预测结果:
[('Xi', 'noun'), ('Jinping', 'verb'), (',', 'noun'), ('male', 'noun'), (',', 'noun'), ('Han', 'noun'), ('ethnicity', '
noun'), (',', 'noun'), ('was', 'noun'), ('born', 'verb'), ('in', 'noun'), ('June', 'verb'), ('1953', 'noun'), ('and',
'noun'), ('is', 'noun'), ('from', 'noun'), ('Fuping', 'noun'), (',', 'noun'), ('Shaanxi', 'punct'), ('Province', 'verb
'), ('.', 'noun'), ('He', 'noun'), ('began', 'noun'), ('his', 'noun'), ('first', 'verb'), ('job', 'noun'), ('in', 'pun
ct'), ('January', 'verb'), ('1969', 'noun'), ('and', 'punct'), ('joined', 'verb'), ('the', 'noun'), ('Communist', 'pun
ct'), ('Party', 'verb'), ('of', 'noun'), ('China', 'punct'), ('(', 'verb'), ('CPC', 'noun'), (')', 'punct'), ('in', 'v
erb'), ('January', 'verb'), ('1974', 'noun'), ('.', 'punct'), ('Xi', 'punct'), ('graduated', 'punct'), ('from', 'punct
'), ('School', 'punct'), ('of', 'punct')]
```

实验总结：

（1）现在的中英文分词效果较好且性能强大，如 jieba 这样的分词工具调用方便好用

（2）Bi-LSTM 结合 CRF 进行词性标注理论上效果良好。在上述实验中中文词性标注效果较好但是英文效果较差，推测是因为英文训练集数据量过小。

附源代码

#cn.py

```
# -*- coding: utf-8 -*-
import jieba.posseg as pseg
import stanza
from snownlp import SnowNLP
import thulac
import pynlpir

def chinese_pos_tagging(file_path):
    # 读取文件
    with open(file_path, 'r', encoding='utf-8') as f:
        text = f.read().strip()

    # 1. Jieba 词性标注
    jieba_tags = pseg.cut(text)
    print("\n=== Jieba 标注结果 ===")
    for word, pos in jieba_tags:
        print(f"{word}/{pos}", end=" ")

    # 2. Stanza 词性标注
    nlp = stanza.Pipeline('zh', processors='tokenize,pos',
download_method=None, proxies=proxies)
    doc = nlp(text)
    print("\n\n=== Stanza 标注结果 ===")
    for sent in doc.sentences:
        for word in sent.words:
            print(f"{word.text}/{word.upos}", end=" ")

    # 3. SnowNLP 分词（无词性标注）
    s = SnowNLP(text)
    print("\n\n=== SnowNLP 分词结果 ===")
    print(list(s.tags))
    # 4. THULAC 词性标注
    thu = thulac.thulac(seg_only=False)
    thu_tags = thu.cut(text)
    print("\n\n=== THULAC 标注结果 ===")
    for word, pos in thu_tags:
        print(f"{word}/{pos}", end=" ")

    # 5. NLPIR 词性标注
    pynlpir.open()
    nlpir_tags = pynlpir.segment(text)
    print("\n\n=== NLPIR 标注结果 ===")
```

```
    for word, pos in nlpir_tags:
        print(f"{word}/{pos}", end=" ")
    pynlpir.close()


if __name__ == "__main__":
    proxies = {
    "http": "http://127.0.0.1:7890",
    "https": "http://127.0.0.1:7890"
    }

# 下载模型时指定代理
    chinese_pos_tagging("Chinese.txt")
```

#en.py

```
import nltk
import spacy
import stanza
proxies = {
    "http": "http://127.0.0.1:7890",
    "https": "http://127.0.0.1:7890"
}

def english_pos_tagging(file_path):
    # NLTK 实现
    with open(file_path, 'r', encoding='utf-8') as f:
        text = f.read()
    tokens = nltk.word_tokenize(text)
    nltk_tags = nltk.pos_tag(tokens)
    print("NLTK 标注结果:", nltk_tags)

    # SpaCy 实现
    nlp = spacy.load("en_core_web_sm")
    doc = nlp(text)
    spacy_tags = [(token.text, token.pos_) for token in doc]
    print("\nSpaCy 标注结果:", spacy_tags)

    # Stanza 实现
    nlp = stanza.Pipeline(lang="en",
processors="tokenize,pos",  download_method=None, proxies=proxies)  # 仅
启用分词和词性标注
    doc = nlp(text)
    stanford_tags = [
        (word.text, word.upos)
```

```
        for sentence in doc.sentences
        for word in sentence.words
    ]
    print("\nStanza 标注结果:", stanford_tags)

english_pos_tagging("English.txt")
```

#CRF 训练 源代码：

```
# -*- coding: utf-8 -*-
#环境：tf
import numpy as np
from sklearn.model_selection import train_test_split
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Embedding, Bidirectional, LSTM,
Dense
import tensorflow as tf
import tensorflow_addons as tfa
import re

def parse_data(file_path):
    sentences = []
    tags = []
    with open(file_path, 'r', encoding='gbk') as f:
        for line in f:
            line = line.strip()
            if not line:
                continue

            parts = line.split()
            if not parts:
                continue

            words = []
            word_tags = []
            for part in parts[1:]:

                match = re.match(r'^(.+)/([a-zA-Z]+)$', part)
                if match:
                    word = match.group(1)
```

```
                    tag = match.group(2).lower()
                    words.append(word)
                    word_tags.append(tag)

            if words and word_tags:
                sentences.append(words)
                tags.append(word_tags)

    return sentences, tags


def conllu_to_sentences_tags(conllu_file):
    sentences = []
    tags = []
    current_sentence = []
    current_tags = []

    with open(conllu_file, 'r', encoding='utf-8') as f:
        for line in f:
            line = line.strip()

            if line.startswith('#') or not line:

                if line.startswith('# sent_id') and current_sentence:
                    sentences.append(current_sentence)
                    tags.append(current_tags)
                    current_sentence = []
                    current_tags = []
                continue


            fields = line.split('\t')
            if len(fields) >= 4:
                word = fields[1]
                pos_tag = fields[3]

                current_sentence.append(word)
                current_tags.append(pos_tag.lower())

    if current_sentence:
        sentences.append(current_sentence)
        tags.append(current_tags)

    return sentences, tags
```

```
# sentences, tags = parse_data('dataset.txt')
conllu_file = 'enronsent00_01.xml.conllu'
sentences, tags = conllu_to_sentences_tags(conllu_file)


words = list(set(word for sent in sentences for word in sent))
tags_flat = ["PAD"] + list(set(tag for tag_seq in tags for tag in tag_seq))
word2idx = {w: i + 1 for i, w in enumerate(words)}
word2idx["PAD"] = 0
word2idx["UNK"] = len(word2idx)
tag2idx = {t: i for i, t in enumerate(tags_flat)}
idx2tag = {i: t for t, i in tag2idx.items()}

sent_lengths = [len(sent) for sent in sentences]
max_len = np.max(sent_lengths)
mean_len = np.mean(sent_lengths)
num_tags = len(tag2idx)

X = [[word2idx.get(w, word2idx["UNK"]) for w in s] for s in sentences]
X = pad_sequences(X, maxlen=max_len, padding='post',
value=word2idx["PAD"])

y = [[tag2idx[t] for t in ts] for ts in tags]
y = pad_sequences(y, maxlen=max_len, padding='post', value=tag2idx["PAD"])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)


input_layer = Input(shape=(max_len,))
embedding = Embedding(input_dim=len(word2idx), output_dim=64,
mask_zero=True)(input_layer)
lstm = Bidirectional(LSTM(64, return_sequences=True))(embedding)
logits = Dense(num_tags)(lstm)

model = Model(inputs=input_layer, outputs=logits)

transition_params = tf.Variable(tf.random.uniform(shape=(num_tags,
num_tags)))

@tf.function
def crf_loss(y_true, y_pred):
    y_true = tf.cast(y_true, dtype=tf.int32)
```

```
    sequence_lengths =
tf.math.reduce_sum(tf.cast(tf.math.not_equal(y_true, 0), tf.int32),
axis=1)
    log_likelihood, _ = tfa.text.crf_log_likelihood(
        inputs=y_pred,
        tag_indices=y_true,
        sequence_lengths=sequence_lengths,
        transition_params=transition_params
    )
    return -tf.reduce_mean(log_likelihood)

model.compile(optimizer='adam', loss=crf_loss)


model.fit(X_train, y_train, batch_size=2, epochs=10)



# test_sent = ['3', '月', '23', '日', '下午', '青年报', '杯赛', 'U19', '邀
请赛', '在', '越南', '芽庄', '进行', '前', '国脚', '曲波',
#              '挂帅', '的', '中国', 'U19B', '队', '迎战', '泰国', 'U19', '
上半场', '国青队', '的', '门户', '大开', '泰国', '在', '第', '11', '分钟', '
和', '第', '17', '分钟', '连进', '2', '球', '半场', '国青', '0', '射门', '0',
'角球', '几乎', '被', '完全', '压制', '下半场', '国青', '的', '进攻', '一度',
'有所', '起色', '并', '由', '马辅', '渔', '利用', '远射', '扳回', '一', '球
', '但', '最终', '未能', '扳平', '比分', '全场', '比赛', '结束', '国青',
#            '1', '2', '输球', '继', '中国', '杯国足', '0', '1', '输给', '
泰国', '之后', '3', '天内', '遭遇', '泰国', '足球', '双杀']

test_sent = ['Xi', 'Jinping', ',', 'male', ',', 'Han', 'ethnicity', ',',
'was', 'born', 'in', 'June', '1953', 'and', 'is', 'from', 'Fuping', ',',
'Shaanxi', 'Province', '.' ,
 'He', 'began', 'his', 'first', 'job', 'in', 'January', '1969', 'and',
'joined', 'the', 'Communist', 'Party', 'of', 'China', '(', 'CPC', ')', 'in',
'January', '1974', '.',
 'Xi', 'graduated', 'from', 'School', 'of', 'Humanities', 'and', 'Social',
'Sciences', ',', 'Tsinghua', 'University', 'where', 'he', 'completed', 'an',
'in-service', 'graduate', 'program', 'in', 'Marxist', 'theory', 'and',
'ideological', 'and', 'political', 'education', '.',
'He', 'holds', 'a', 'Doctor', 'of', 'Law', 'degree', '.','Xi', 'is',
'currently', 'General', 'Secretary', 'of', 'the', 'CPC', 'Central',
'Committee', ',', 'Chairman', 'of', 'the', 'CPC', 'Central', 'Military',
'Commission', ',', 'President', 'of', 'the', 'People', "'s", 'Republic',
'of', 'China', '(', 'PRC', ')', ',', 'and', 'Chairman', 'of', 'the', 'PRC',
'Central', 'Military', 'Commission', '.']
```

```
test_seq = [[word2idx.get(w, word2idx["UNK"]) for w in test_sent]]
test_pad = pad_sequences(
    test_seq,
    maxlen=max_len,
    padding='post',
    value=word2idx["PAD"]
)

logits_val = model.predict(test_pad)

sequence_lengths = tf.math.reduce_sum(
    tf.cast(tf.math.not_equal(test_pad, word2idx["PAD"]), tf.int32),
    axis=1
)

sequence_lengths = tf.minimum(sequence_lengths, max_len)

viterbi_seq, _ = tfa.text.crf_decode(
    logits_val,
    transition_params,
    sequence_lengths
)

pred_ids = viterbi_seq.numpy()[0][:len(test_sent)]
pred_tags = [idx2tag[i] for i in pred_ids]

print("\n 预测结果：")
print(list(zip(test_sent, pred_tags)))
```